

Introduction & Hook

Ask:

“Why can’t you push through a wall in advanced VR simulations?”

Explain:

Because the system applies **force feedback**, creating resistance.

Unlike vibration (which only signals touch), force feedback **physically resists movement**.

What is Force Feedback?

Basic Concept

Definition:

Force feedback is a type of haptic feedback that applies physical forces to the user to simulate resistance, weight, or impact.

Examples:

- Steering wheel resistance in driving simulators
 - VR gloves resisting finger movement
 - Surgical tools providing pressure
-

Key Difference:

Vibrotactile	Force Feedback
Vibration only	Physical resistance
Low realism	High realism

Important Insight:

Force feedback simulates *interaction with physical objects*, not just contact.

Haptic Devices

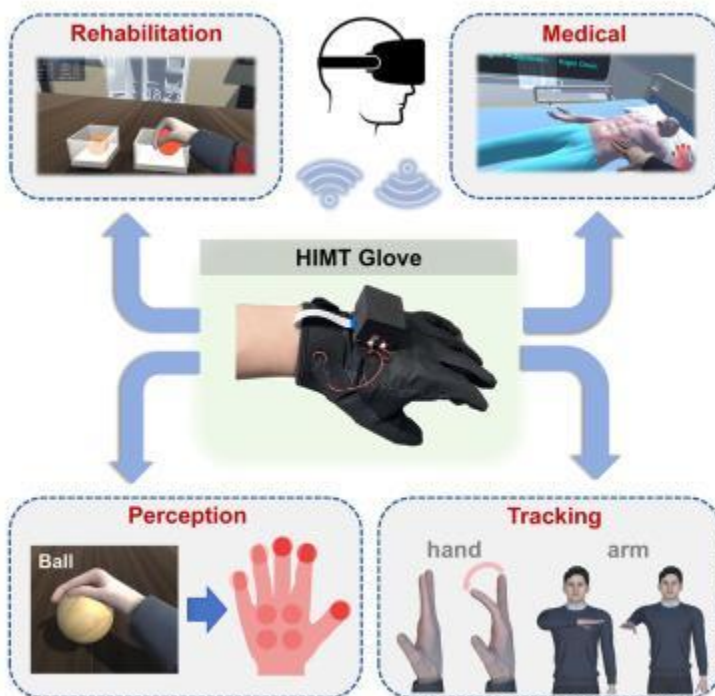
Types of Haptic Devices

1. Haptic Controllers



- Provide vibration + limited force
 - Used in gaming and general VR
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2. Haptic Gloves



- Track finger movement
- Provide resistance to fingers

- Used in:
 - o Medical training
 - o Industrial VR
-

3. Force Feedback Devices (Robotic)

- Robotic arms or stylus devices
 - Provide precise force control
 - Used in:
 - o Surgery simulation
 - o Research labs
-

Structure of Force Feedback Systems

System Architecture

Explain the structure step-by-step:

Basic Flow:

```
User Action
↓
Sensors detect movement
↓
VR System processes interaction
↓
Force calculation (physics engine)
↓
Actuator applies force
↓
User feels resistance
```

Components of Force Feedback System

1. Sensors

- Detect position, movement, and force
 - Examples:
 - IMU
 - Position sensors
-

2. Controller (Processing Unit)

- Runs physics simulation
 - Calculates forces
-

3. Actuators

- Apply force to user
 - Types:
 - Motors
 - Pneumatic systems
 - Electromagnetic actuators
-

4. Feedback Loop

System continuously updates:

- Input → Processing → Output → Repeat
-

Key Concept:

Force feedback systems operate in a real-time loop.

Force Feedback + Physics Integration

Connection with Physics Engine

Explain:

Force feedback depends on:

- Collision detection
 - Object properties (mass, stiffness)
 - Physics simulation
-

Example:

User presses virtual wall:

- Collision detected
 - Physics engine calculates force
 - Device applies resistance
-

Simple Formula (Hookes law):

Hooke's Law:

The force applied by a spring is directly proportional to its displacement from the equilibrium position.

Mathematically: $F = kx$, where k is stiffness and x is displacement.

$$F=kx$$

Where:

- F = force
 - k = stiffness
 - x = displacement
-

Real VR Example

VR surgery simulation:

- User inserts tool
- Feels resistance from tissue
- Adjusts pressure accordingly

This improves:

- Skill training
- Precision
- Realism



58–60 min → Recap & Questions

Key Points:

- Force feedback provides resistance and physical sensation
- Haptic devices include controllers, gloves, and robotic systems
- Force feedback systems use sensors, processors, and actuators
- Physics engine calculates forces

Quick Questions:

1. What is force feedback?
 2. Difference between vibration and force feedback?
 3. What are actuators?
 4. Why is physics important in force feedback?
-

1 Numerical: Basic Force Feedback (Hooke's Law)

Formula

$$F = kx$$

Where:

- F = force
 - k = stiffness
 - x = displacement
-

Problem

A VR haptic device simulates a virtual spring with stiffness $k = 200 \text{ N/m}$.

User presses the object by 0.01 m .

Find the force applied.

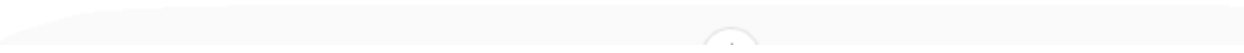
Solution

$$F = 200 \times 0.01 = 2 \text{ N}$$

✓ Force = 2 N

Teaching Point

This is how VR simulates resistance when pressing objects.



2 Numerical: Increased Resistance

Problem

A virtual wall has stiffness $k = 500 \text{ N/m}$.

User pushes it by 0.02 m .

Find force applied.

Solution

$$F = 500 \times 0.02 = 10 \text{ N}$$

✓ Force = 10 N

Insight

Higher stiffness → stronger resistance → harder surface

3 Numerical: Comparing Soft vs Hard Object

Problem

Two virtual objects:

- Object A: $k = 100\text{N/m}$
- Object B: $k = 400\text{N/m}$

Displacement = **0.01 m**

Find forces and identify which feels harder.

Solution

Object A:

$$F = 100 \times 0.01 = 1\text{N}$$

Object B:

$$F = 400 \times 0.01 = 4\text{N}$$

Conclusion

- Object B produces higher force
- Feels harder

4 Numerical: Force from Mass and Acceleration

Formula

$$F = m \times a$$

Problem

A virtual object of mass **3 kg** accelerates at **2 m/s²** during interaction.

Find force applied.

Solution

$$F = 3 \times 2 = 6N$$

✓ Force = 6 N

Teaching Point

Physics engines calculate forces like this in VR.

5 Numerical: Work Done in Force Feedback

Formula

$$W = F \times d$$

Problem

A haptic device applies 5 N force over displacement of 0.02 m.

Find work done.

Solution

$$W = 5 \times 0.02 = 0.1 J$$

✓ Work done = 0.1 Joules

6 Numerical: Force Feedback Loop Timing

Problem

A VR system updates force feedback at 1000 Hz.

Find time per update.

Solution

$$Time = \frac{1}{1000} = 0.001 \text{ sec}$$

✓ Time per update = 1 ms

Teaching Insight

Haptic systems require very fast updates for smooth feedback.

7 Numerical: Maximum Force Limit

Problem

A haptic device can apply maximum force of 8 N.

Calculated force is 12 N.

What will system do?

Solution

Since:

$$12N > 8N$$

Device cannot apply full force.

✓ Output is limited to 8 N

Teaching Point

Real devices have physical limits.

8 Numerical: Energy Stored in Virtual Spring

Formula

$$E = \frac{1}{2}kx^2$$

Problem

A virtual spring has:

- $k = 300N/m$
- $x = 0.02m$

Find stored energy.

Solution

$$E = \frac{1}{2} \times 300 \times (0.02)^2$$

$$E = 150 \times 0.0004 = 0.06J$$

✓ Energy = 0.06 Joules